

Life Cycle Assessment Integrated Machine Learning for Structural Exploration

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Abstract

Life cycle assessment (LCA) is rarely integrated in the early design phases, where it would be most effective. This article proposes a machine learning framework to bridge the gap between structural exploration and multi-criteria environmental optimization. Using a dataset of over 90,000 parametrically generated residential buildings, we first compare supervised learning models to establish a robust surrogate model capable of predicting LCA indicators (ReCiPe 2016) in real time; a multilayer perceptron (MLP) is selected for its superior performance. We then address the inverse design problem using a conditional variational autoencoder (CVAE). This generative model is trained to propose structural configurations that simultaneously satisfy eight distinct environmental constraints, thus going beyond single-objective optimization. We evaluate the model's performance using Pareto front analysis and hypervolume metrics, demonstrating its ability to generate valid and efficient designs. Finally, we analyze the latent space to visualize the critical trade-offs between material choices, geometry, and environmental impacts, thus providing an innovative tool for sustainable, informed, and interactive design.

Keywords: Generative Design, Life Cycle Assessment (LCA), Conditional Variational Autoencoder (CVAE), Surrogate Modeling, Structural Optimization, Multi-criteria Decision Making

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1 Introduction

The construction and building sector is a major contributor to the current environmental crisis. Indeed, it is responsible for an estimated 37% of global energy-related CO₂ emissions, for about 50% of all material extractions, and it generates over 35% of the European Union's total waste. At the heart of this sector, the engineer is tasked with finding technical solutions to adapt an idea to mechanical, regulatory, and environmental criteria. Through iterative work, the object, born from intuition, converges toward a version that validates these criteria. The drawback of this approach is that the initial choices are responsible for most of the impact on all criteria [2], yet they are often made intuitively. Generative design seeks to overturn this paradigm and rationalize the process of form finding. Once the problem is clearly defined, diverse optimization techniques, recently enhanced by machine learning, are implemented to explore large solution spaces. While generative design optimizes the quantity of material very well, it poorly manages the choice of end of life or logistical provenance, often overlooking impacts during manufacturing.

1.1 Integrating LCA to generative design

1.1.1 Life Cycle Assessment methodology : ReCiPe 2016

To mitigate this, Life Cycle Assessment (LCA) acts as an essential constraint. LCA is a standardized assessment method [2] that quantifies the potential environmental impacts of a system throughout its entire life cycle. Through its comprehensive, multi-criteria approach, it identifies the most polluting stages and prevents pollution transfer. In France, LCA has become widespread due to the RE2020 regulations, forcing the entire supply chain to produce Environmental Product Declarations (EPDs). Yet, for most civil-engineering projects, implementing a full LCA at an early stage remains too complex and expensive. "Proxies" are often used instead of a true multi-criteria LCA, and the method is typically applied after the fact to validate a decision rather than upstream for design.

The ReCiPe 2016 Method is a Life Cycle Assessment method that converts the physical flows of the inventory into environmental indicators at two complementary levels: robust midpoints and aggregated endpoints. The Midpoint indicators are located early in the cause-and-effect chain and translate the inventory flows into 18 specific environmental problems. Their main characteristic is their scientific robustness and low uncertainty: they quantify the potential impact without extrapolating to ultimate consequences, making them ideal for precise technical eco-design, although they can sometimes be difficult for a non-expert decision-maker to interpret comparatively.

- Global warming (GWP): $kg\ CO_2\ eq$
- Ozone depletion (ODP): $kg\ CFC-11\ eq$
- Ionizing radiation (IRP): $kBq\ Co-60\ eq$
- Photochemical oxidant formation - human health (HOFPP): $kg\ NO_x\ eq$
- Photochemical oxidant formation - ecosystems (EOFP): $kg\ NO_x\ eq$
- Particulate matter formation (PMFP): $kg\ PM_{2.5}\ eq$
- Freshwater ecotoxicity (FETP): $kg\ 1,4-DCB\ eq$

- Marine ecotoxicity (METP): kg 1,4-DCB eq
- Terrestrial ecotoxicity (TETP): kg 1,4-DCB eq
- Human carcinogenic toxicity (HTPc): kg 1,4-DCB eq
- Human non-carcinogenic toxicity (HTPnc): kg 1,4-DCB eq
- Land use (LOP): $m^2 \cdot a$ crop eq
- Terrestrial acidification (TAP): kg SO₂ eq
- Freshwater eutrophication (FEP): kg P eq
- Marine eutrophication (MEP): kg N eq
- Water consumption (WCP): m^3
- Mineral resource scarcity (SOP): kg Cu eq
- Fossil resource scarcity (FFP): kg oil eq

The Endpoint indicators model the end of the impact chain by linearly aggregating the previous indicators into three protection areas: Human Health (DALYs), Ecosystem Quality (species per year), and Resource Depletion (USD). Their strength lies in their ease of interpretation for strategic decision-making (simplified arbitration); however, this synthesis comes at the cost of high uncertainty due to the many complex assumptions needed to convert a physico-chemical phenomenon into concrete damage to society or nature.

- Human Health: *DALY* (Disability-Adjusted Life Years)
- Ecosystem Quality: *species · yr* (Local species loss over time)
- Resource Scarcity: *USD₂₀₁₃* (Surplus cost)

1.1.2 State of the art for Generative Design with LCA

It is established that moving from intuitive design to AI-assisted design can improve the energy and environmental performance of a construction project [1]. Since the popularization of the LCA methodology, particularly since it was made mandatory in France with the RE2020 regulation, a number of studies have linked this methodology to optimization and machine learning techniques. The aim is to make the LCA process more accessible by allowing architects to assess in real time the ecological consequences of their structural choices from the first sketches. [6] This work is in particular part of the continuation of research carried out on the use of variational autoencoders to generate structures under LCA constraints. [7].

1.2 Objectives of this study

This work seeks to link generative design assisted by supervised learning methods and life cycle assessment (LCA). Firstly, it aims to enable the real-time estimation of LCA values for parameterized structures. This should allow designers to make structural choices while being aware of their environmental impacts (**Direct Problem**). Secondly, the objective will be to address the **Inverse Problem**. Given a number of LCA (and structural) constraints, is it possible to generate diverse structures that meet the given targets? This work follows on from the work carried out by M.Pollet et al. [7], in which a CVAE architecture was proposed to address the inverse problem under the constraint of a single LCA indicator. Beyond the various ways of assisting

design with LCA methodology, this work also proposes methodological elements for integrating the specificities of life cycle assessment into deep learning, as well as for processing and analyzing the results.

1.3 Generating a database of structures

Currently, there is no sufficiently large real-world database for machine learning that combines structural representation with indicators using a LCA methodology. Therefore, in this study, the choice was made to use a parametrically generated database [5]. This database was designed to focus on multi-story housing buildings, with a rectangular floor area. The structural system of the buildings consists of floors supported by a network of beams and columns, with one or more shear wall cores. A total of 100,000 designs were sampled from the design space using a Sobol sequence. Of these 100,000 buildings, 7,203 variants did not meet the mechanical sizing criteria, bringing the effective number of rows in the database to 92,797 entries.

Design parameter	Possible values
Building width (m)	{6, 9, 12, 15}
Building length (m)	{12, 15, . . . , 87, 90}
Core(s) width (m)	{3, 6}
Core(s) length (m)	{3, 6}
Cores spacing (m)	{6, 9, . . . , 24, 27}
Core(s) lateral position (% of building width)	[0, 100]
Number of floors	{3, 4, 5, 6, 7, 8}
Columns longitudinal spacing (m)	{3, 6, 9}
Columns lateral spacing (m)	{3, 6, 9}
Floors material	{CLT, Steel-Concrete composite, Concrete}
Beams material	{Glulam, Steel, Concrete}
Columns material	{Glulam, Steel, Concrete}

Table 1 Parametric variables describing the database structures

2 From Simulation to Prediction: Supervised Learning Surrogate

2.1 Preparation and Validation

This section aims to test machine learning (ML) models for predicting the life cycle assessment (LCA) indicators of parameterized buildings. The goals are : to train classical machine learning models to predict their environmental impacts, to use various types of models (linear and polynomial regression, decision trees, simple multilayer neural networks, etc.) and compare their performances, and to predict environmental impacts according to all LCA indicators. This kind of model, which addresses the direct problem of predicting the LCA parameters of given structures, is referred to as a surrogate model.

Performance indicators were normalized by building surface area to obtain intensive quantities (expressed per unit area). This transformation aims to make observations comparable regardless of their size and to reduce variance due to scale effects. Among all the indicators in the ReCiPe 2016 methodology, we chose to predict the midpoint indicators. Z-scaling transforms the features so they have a mean (μ) of 0 and a standard deviation (σ) of 1. This is crucial for algorithms sensitive to the magnitude of data (e.g., Gradient Descent, SVM). The formula for a single observation x is:

$$z = \frac{x - \mu}{\sigma}$$

For variables representing design characteristics, we consider different types of variables: continuous and discrete. Continuous variables are also Z-scaled, while discrete variables are processed using One-Hot encoding.

2.1.1 Supervised Learning Models

To create this prediction model, we will compare the performance of different types of learning models exhibiting different characteristics of flexibility and interpretability. Flexibility refers to a model's ability to adapt to various forms of data and capture complex relationships. Interpretability refers to the degree to which a given model is understandable to humans. In the remainder of this study, to validate our objectives, we will explore various trade-offs in the Interpretability-Flexibility framework. Specifically, we will seek to implement simple and interpretable models such as linear regression, as well as highly flexible and less interpretable models like multilayer neural networks. The models we will test are linear regression (LR), random forest (RF), support vector regression (SVR), and multilayer perceptrons (MLP)

A central concept in supervised learning is the tradeoff between bias and variance. Bias represents the error due to overly simplistic assumptions in the learning algorithm (underfitting). Variance, on the other hand, represents the error due to the model being too sensitive to fluctuations in the data (overfitting). One of the challenges in developing a good supervised learning model is finding this tradeoff between bias and variance. This allows us to extract characteristics from the training data without simply copying them.

In this work, both Single-Output and Multi-Output models were tested. The Single-Output architecture has the advantage of preventing noise from one indicator from affecting other predictions. However, correlations between indicators are learned more effectively. In Deep Learning, the first layers of the network learn common features are usefull for all tasks, which often improves overall performance. This is why we favored a Multi-Output architecture for neural networks.

Linear Regression

It is a simple and interpretable way to define a baseline. For any predicted LCA indicator, the linear coefficients have significance and can be interpreted. However, it does not capture non-linear relationships. It is also sensitive to collinearity and extreme values. For example $floor_area = n_floors \times building_width \times building_height$. To avoid collinearity, the floor area can be removed from the database and recomputed afterward. Linear Regression relies on a constant variance of errors (homoscedasticity).

Random Forest (RF)

It is a set of decision trees, each trained on a random sample. It captures non-linear relationships and variable interactions with a low risk of overfitting. It handles categorical variables and different orders of magnitude very well. The downside is that random forests (RF) do not predict well beyond the range of the training data. Random Forests handle discrete variables effectively, explaining their strong performance on this database.

Support Vector Regressor (SVR)

It looks for a function that approximates the data within a certain range (ε), minimizing errors outside this range while keeping the model as flat as possible. Often uses a Radial Basis Function kernel to model nonlinearity. It has very satisfactory performance for nonlinear relationships with little noise and precise control of the bias–variance tradeoff via hyperparameters (C, ε, γ). It is less interpretable and more sensitive to the choice of kernel and parameters. Yet, an SVR can capture complex non-linear relations and yield satisfactory results for training such a surrogate model.

Multi-Layer Neural Network (MLP – Multi-Layer Perceptron)

A MLP is a neural network with several hidden layers that learns complex nonlinear representations of inputs. It is a highly flexible model, capable of approximating almost any function, performing well with large amounts of data and proper preprocessing. It requires large amounts of data and proper scaling. There is a risk of overfitting and low interpretability. A MLP is suitable for highly nonlinear problems and advanced modeling of interactions between variables.

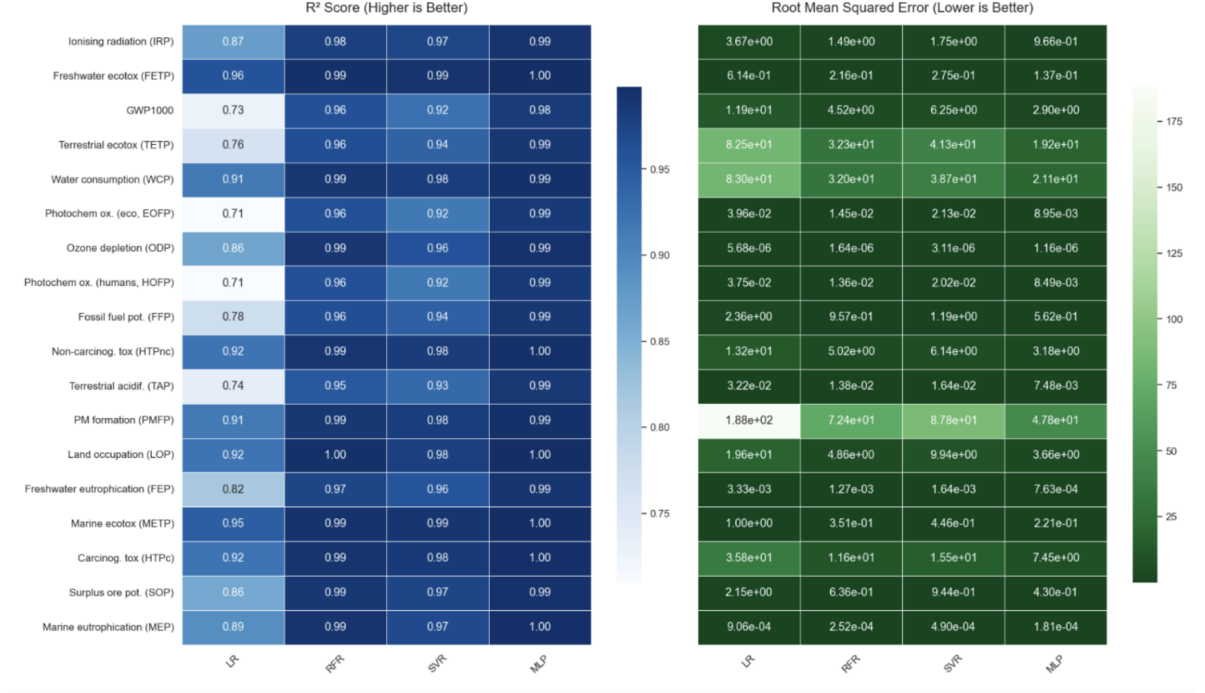


Fig. 1 Comparison of results between different supervised learning models on all midpoint indicators of the ReCiPe 2016 method, on the validation database.

2.2 Performance of the models

2.2.1 Metrics

To assess the predictive performance of the models for each output indicator, two metrics are considered: the coefficient of determination (R^2) and the Root Mean Squared Error (RMSE). These metrics capture different aspects of model accuracy.

$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$. The R^2 score measures the proportion of the variance in the observed data that is explained by the model, where y_i are the observed values, $\hat{y}_i$ are the predicted values, and $\bar{y}$ is the mean of observations.

The $\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}$ is the mean squared error. The square inside the expression tends to amplify the effect of outliers.

2.2.2 Results

MLP significantly outperforms linear and kernel approaches (SVR). It captures non-linearities that the linear model (LR) is mathematically unable to see [1](#). Although it achieves the best absolute scores, the difference with Random Forest (RFR) is not always statistically significant across all indicators, suggesting that these two non-linear methods reach the theoretical performance ceiling on this dataset. With

an average error of approximately 1.8% across all indicators, taking into account differences in magnitude, the MLP model is retained for the remainder of this study.

3 Generative deep learning model for structural exploration : The CVAE Framework

3.1 Mathematical Formulation of the LCA-Informed CVAE

Variational AutoEncoders (VAEs) are generative models that learn to encode and decode data in a latent space. They are capable of generating new data similar to the training set. VAEs were originally introduced by Diederik P. Kingma and Max Welling in 2013 in a seminal paper titled "Auto-Encoding Variational Bayes." [3] In this publication, they impose a continuous structure on the latent space. The data is compressed using Gaussian probability distributions, according to the variational inference process. The reparameterization trick $z = \mu + \sigma \cdot \epsilon$ makes random sampling differentiable for backpropagation. VAEs are notably used for image generation, data denoising, anomaly detection, and data compression. The Conditional Variational Encoding Model (CVAE) is an extension of the VAE that guides the generation process using additional information called a condition. The condition specifies to the model what it should generate. The condition is merged in the encoder with the latent vector in the decoder. Therefore, using CVAE requires labeled data for training. CVAE was introduced by Kihyuk Sohn, Honglak Lee, and Xinchen Yan in 2015. [8]

3.1.1 Model Architecture

The proposed framework utilizes a Conditional Variational Autoencoder (CVAE) adapted to act both as a surrogate model for environmental impact prediction and as a generative model. In the following section, x denotes the input building design vector, y denotes the target performance vector (LCA indicators), and z denotes the latent vector. The architecture consists of three distinct neural networks. The Probabilistic Encoder (q_ϕ) maps the design x to an LCA performance $\hat{y}$ and the parameters of the approximate posterior distribution of the latent space: $q_\phi(z|x) = \mathcal{N}(\mu_q(x), \sigma_q^2(x))$. The Conditional Prior Network (p_ψ) maps the target performance y to the parameters of the prior distribution of the latent space. This structures the latent space explicitly with respect to y .

$$p_\psi(z|y) = \mathcal{N}(\mu_p(y), \sigma_p^2(y))$$

Finally, the Decoder (p_θ) maps the concatenated target performance y and latent vector z to a reconstructed design $\hat{x} \sim p_\theta(x|y, z)$

3.1.2 Loss functions

During training, the parameters ϕ , ψ , and θ are optimized to minimize a composite loss function $\mathcal{L} = \mathcal{L}_{surrogate} + \mathcal{L}_{reconstruction} + \mathcal{L}_{KL}$. Where:

- $\mathcal{L}_{surrogate}$: Measures the prediction error on y (MSE).

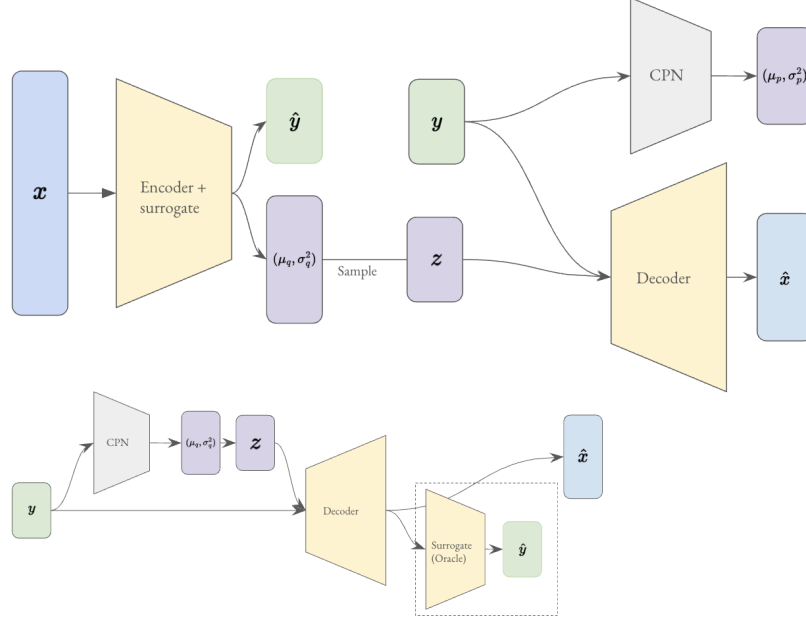


Fig. 2 CVAE architecture, used in this work, for training and generation.

- $\mathcal{L}_{reconstruction}$: Measures the reconstruction error on x . It is a composite term that sums the Mean Squared Error (MSE) for continuous variables and the Binary Cross-Entropy (BCE) for categorical/discrete variables.
- $\mathcal{L}_{KL}$: The Kullback-Leibler divergence ensures that the posterior aligns with the conditional prior $\mathcal{L}_{KL} = D_{KL}(q_{\phi}(z|x) \parallel p_{\psi}(z|y))$.

3.1.3 Training a CVAE

During training, for each model, multiplicative factors are randomly drawn and associated with each term of the loss function. This can have a significant effect on the model's training and lead to some of the erratic behaviors.

Temperature (T) and the KL parameter (C) control the balance between the model's "creativity" and its accuracy, as well as the topological organization of the latent space. Temperature manages variance during generation by acting as a noise multiplier in the Reparametrization Trick:

$$z = \mu + T \cdot (\sigma \cdot \epsilon)$$

. With temperatures below 1, the model selects points very close to the mean μ . The generations are accurate but not very varied. With $T > 1$ (High temperature), the model explores further into the distribution. This allows for more diverse generations, but with an increased risk of artifacts. The introduction of noise significantly reduces

the validation loss. The KL parameter C serves as a regularization for the KL divergence loss. If the parameter is too small: The model acts as a classic autoencoder. It reconstructs very well, but the latent space is "messy." The generation of new data will be poor. If the parameter is too strong: The model focuses too much on having a perfect distribution and ignores the input data. This is the phenomenon of "Posterior Collapse." The generation ignores the conditions.

KL annealing, or warm-up, is a training technique that allows the model to learn to reconstruct faithfully before introducing noise. Indeed, at the beginning of training, if a high weight is immediately imposed on the KL, the encoder "gives up" and encodes everything towards zero to satisfy the constraint, without learning to encode the image details. To counteract this, the KL weight is set to 0, the model is allowed to first learn to reconstruct the image well, and then the weight C is gradually increased over time.

In a model subject to posterior collapse, the regularization constraint (KL) becomes unbearable for the model, forcing it to abandon encoding information. Effect: The KL loss quickly falls to 0. The total loss is low, but the reconstruction metrics are poor and stagnate.

In the case of overfitting, the reconstruction is excellent. The latent space is "disconnected" from the prior. The model can no longer generate new coherent structures from y alone.

3.1.4 Inference (Generation)

For the generative task, a target performance y_{target} is specified (LCA-condition). The generation process follows two steps. A latent vector is sampled from the conditional prior: $z \sim p_\psi(z|y_{target})$. This sampled vector is run through the decoder to produce the parametric design: $\hat{x} = p_\theta(x|y_{target}, z)$.

After this, we can use the surrogate to predict the LCA characteristics of the generated solutions. And check how closely they match the LCA-condition.

3.2 Architecture modification

In this work, the CVAE model structure is retained with one modification. The surrogate, present in the architecture of [7], is replaced by a multilayer neural network trained in Section 2. Indeed, depending on the model, surrogates, while performing the same task of predicting LCA indicators from building characteristics, exhibit highly variable performance. Even though the surrogate's role is not central to the CVAE architecture, it allows for a rapid evaluation of the performance of the generated solutions. Therefore, by using the same neural network, we obtain a stable performance evaluation of the generated structures, which is a major advantage for comparing models.

4 Scaling to Multi-Criteria Design

4.1 The Challenge of High-Dimensional Performance Spaces

One of the main contributions of this study is to examine the challenges associated with adding LCA indicators to the model. This work goes beyond the model proposed by [7], where only GWP values were generated, making it easy to compare the performance of different models. Extending the use of CVAE to other LCA indicators raises the question of evaluating model performance. Several criteria can be considered. The model’s ability to generate structures that meet the required LCA criteria. The diversity of valid structures generated—that is, whether the model can find multiple valid solutions to the same LCA problem. The organization of the latent space.

From a practical standpoint, adding indicators also requires revising the presentation of the results. Using results in 17 dimensions, which corresponds to the number of midpoint indicators, demands clarity in the visual language.

4.1.1 Selection of a subset of indicators

To simplify model exploration and experiment with ways to compare and visualize the data, we chose to limit the training to 8 indicators. This allows us to test the functionality of the architecture proposed by [7] without making the visuals cumbersome. The selection of 8 indicators was based solely on the LCA indicator correlation matrix, ensuring that positively, negatively, and uncorrelated indicators were included. No physical or methodological arguments related to LCA justify selecting one subset of indicators over another. Therefore, it is important to keep in mind that the presented results should be extended to all other indicators to rigorously adhere to the LCA methodology. The selected indicators are:

- GWP: Global Warming Potential
- LOP: Land Occupation Potential
- WCP: Water Consumption Potential
- SOP: Surplus Ore Potential
- ODP: Ozone Depletion Potential
- MEP: Marine Eutrophication Potential
- FFP: Fossil Fuel Potential
- TETP: Terrestrial Eco-Toxicity Potential

4.2 Model Evaluation

4.2.1 Target identification on the Pareto front

As mentioned, one of the challenges of model generation is presenting the model with a well-defined LCA-Structure problem. How do we choose the targets to set for testing the models and obtain results that exceed the performance of the training dataset? One approach is to continuously adjust one indicator and set others at median values, then observe how the model manages to generate structures under these conditions. The major drawback of this methodology is that the resulting LCA targets can lead

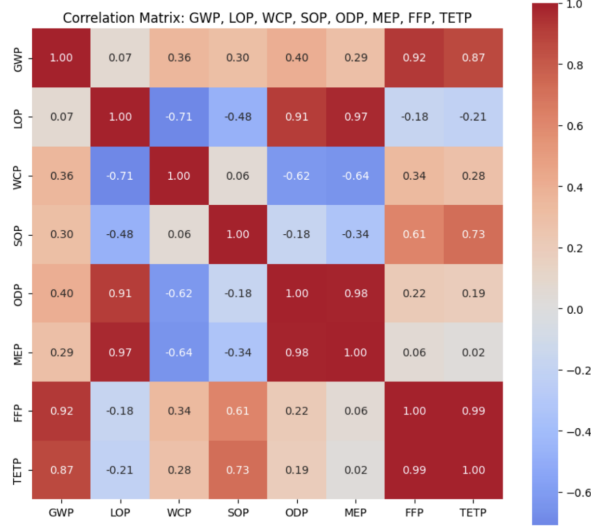


Fig. 3 Correlation matrix of the 8 selected LCA indicators

to unrealizable combinations. It is essential to decouple the physical realizability of the targets from the performance of the models.

To test models on a wide variety of structures with both feasible and high-performing LCA criteria, targets are selected from the Pareto front of the training dataset. The surface-normalized LCA values of the structures in the training dataset form a point cloud in a multidimensional space. The points in this space can be compared using the dominance relationship. A structure S_a is said to be dominated by ($\preceq$) a structure S_b if it has values less than or equal to those of its value on all surface-normalized LCA indicators.

$$S_a \preceq S_b \iff y_a^d \geq y_b^d, \forall d$$

This ordering relationship defines a Pareto front; that is, the set of points that dominate all others. These points represent the structures in the training dataset with the best LCA results. Using the Pareto Peeling method, the points on the Pareto front can be classified into different dominance layers. Within a layer, the points do not dominate one another. They exhibit different trade-offs but similar performance. Conversely, the points in layer 1 dominate those in layer 2 which, in turn, dominate those in subsequent layers. The idea presented in this work is to use the points in the training dataset of the first 10 layers of the Pareto front as targets for generating new structures. This effectively allows for the generation of solutions from demanding, realistic, and diverse compromises. The graph in Figure 4 shows the first 10 Pareto layers for three indicators (GWP, LOP, WCP) obtained using the Pareto Peeling method on the training dataset. These points will serve as targets and points of comparison for the structures generated by the model. The different layers represent equivalent classes according to the partial order between structures.

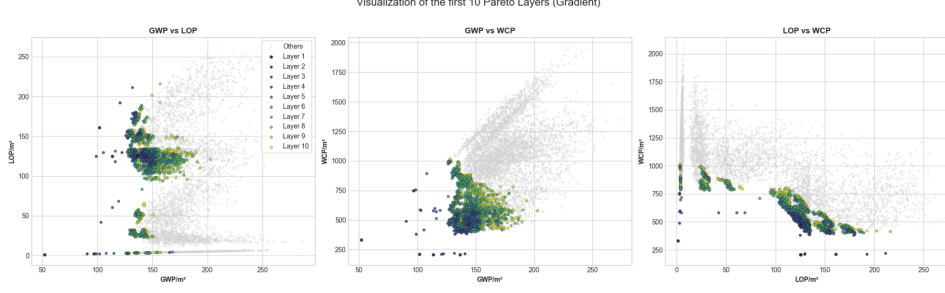


Fig. 4 Visualization of the first 10 Pareto layers for three indicators on the training dataset

4.2.2 Generation of a Pareto approximation set

Thus, for each model m , a number $n_{targets}$ of LCA targets is randomly chosen from the first layers of the Pareto front of the training dataset. Each of these $n_{targets}$ targets is then used to generate p structural solutions. Once generated, these p structures can be ranked according to their satisfaction of the defined LCA target. This post-generation selection process is one of the advantages of the architecture chosen by [7] and significantly improves the overall performance of the model by retaining only the most efficient solutions. In this work, the surrogate initially planned for ranking the structures is replaced by the MLP, whose training was discussed in Section 2, and whose performance is superior and significantly more stable than the surrogates for each of the models. Thus, for a given model m , $n_{targets} \times p$ solutions are generated. These solutions can be represented in the LCA parameter space. Figure 5 shows the $n_{targets} \times p$ structures generated by the model $m = 13$. The set S of generated structures is called a Pareto approximation set.

4.2.3 Hypervolume Indicator

To evaluate the quality of the Pareto approximation set in the 8-dimensional LCA space, we employ hypervolume as a metric for evaluating the models. The hypervolume measures the volume of the objective space that is dominated by the generated solution set S , bounded by a reference point $\mathbf{n}$.

Let $S = \{\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(n_{targets} \times p)}\} \subset \mathbb{R}^d$ be the set of LCA vectors of the generated structures. d is the number of dimensions of the $\mathbf{y}$ LCA vector. Let $\mathbf{n} \in \mathbb{R}^d$ be a reference point chosen such that it is dominated by all valid solutions. We choose the worst (for $\preceq$) in the training dataset, also known as the Nadir. The hypervolume is defined as the Lebesgue measure Λ of the union of hypercuboids formed by each solution and the reference point. It can be normalized by the hypervolume $HV(S^0, \mathbf{r})$ of the training dataset.

$$HV(S, \mathbf{r}) = \Lambda \left(\bigcup_{\mathbf{y} \in S} \{\mathbf{z} \in \mathbb{R}^m \mid \mathbf{y} \preceq \mathbf{z} \preceq \mathbf{r}\} \right) \quad (1)$$

A higher hypervolume indicates a better approximation of the Pareto front, reflecting both superior LCA performance and greater diversity. However, the drawback of this metric is its sensitivity to outliers. To stabilize this metric, one can examine the hypervolume evolution curve as a function of the considered Pareto layers. If a small number of dominant points are responsible for a disproportionate hypervolume measurement, this will be reflected by a jump in the curve between the first layers. In conclusion, analyzing the hypervolume evolution as a function of Pareto layers allows us to verify that hypervolume is a relevant measure.

4.2.4 Generative reliability and diversity metrics

In addition to hypervolume, other custom metrics were considered to evaluate generative performance and the diversity of solutions offered by the different models. We define a *Generative Reliability* metric based on cycle-consistency. Let E_ϕ , D_θ , and P_ψ denote the Encoder, Decoder, and Prior networks, respectively. Given a requested target vector $\mathbf{y}^*$, we generate a set of N candidate structures $\{\hat{\mathbf{x}}_k\}_{k=1}^N$ by sampling from the learned prior: $\mathbf{z}_k \sim P_\psi(\mathbf{z}|\mathbf{y}^*)$, $\hat{\mathbf{x}}_k = D_\theta(\mathbf{z}_k, \mathbf{y}^*)$. Since evaluating the true LCA performance of thousands of generated structures is computationally prohibitive, use a pretrained surrogate model to estimate the realized properties $\hat{\mathbf{y}}_k = E_\phi(\hat{\mathbf{x}}_k)$. The consistency error is calculated as the Mean Absolute Percentage Error (MAPE) between the requested targets and the realized estimates. The final Generative Reliability score R is defined as the success rate over a discretized grid $\mathcal{G}$ of the condition space (spanning the quantiles of the dataset), given a tolerance threshold τ (set to 20%):

$$R = \frac{1}{|\mathcal{G}|} \sum_{\mathbf{y}^* \in \mathcal{G}} \mathbb{I}(\text{MAPE}(\mathbf{y}^*) < \tau) \quad (2)$$

To assess the model’s ability to propose diverse structural solutions for a single environmental target y , we introduce a diversity score. For a fixed condition y , a batch of N designs is generated by sampling latent vectors $z \sim \mathcal{N}(0, I)$. The diversity is quantified as the sum of the variances of each design feature j across the generated batch:

$$\text{Score}_{Div} = \sum_{j=1}^D \text{Var}(x_{gen}^{(j)}) \quad (3)$$

where D represents the total number of design parameters (features of x). A higher score indicates that the model is capable of exploring a wider region of the design space to satisfy the same specific LCA constraints.

4.3 LCA Trade-offs in Latent Space

4.3.1 Using latent space as a visualization tool

Another important question that arises as the number of modeled LCA indicators increases is the interpretability of the model, particularly the interpretability of the projections of the latent space. As mentioned, one of the objectives of generative

design is to assist designers by providing tools to visualize acceptable trade-offs and to avoid unanticipated pollution shifts. Since the CVAE model learns to separate the latent space into different zones, notably according to the LCA indicators, it is natural to seek to visualize how this separation is made and to project these different zones, ultimately resulting in a map of possible LCA trade-offs.

An important question to ask is whether the distortions in the latent space are representative of the correlations between LCA parameters. This involves verifying that similar LCA trade-offs are in regions close to the latent space. The regularity of the latent space, as well as other topological properties, can be important for using the latent space as an interactive visualization or generation tool. Another approach, aimed at maximizing the generative performance of CVAE, involves ignoring the topological and visual characteristics of the latent space. In accordance with the problem posed in the introduction, this work primarily seeks to establish means of assisting human designers in their work through visualization and generation tools. Therefore, the visual aspect of the latent space is considered a significant criterion for model evaluation. While correlated with generative performance, this criterion remains, for the time being, subjective. It is also important to note that the latent space represents only part of the model and that the role of the decoder must also be taken into account.

4.3.2 Choosing a projection method

Principal Component Analysis (PCA) is an orthogonal projection that seeks to find new axes (principal components) that maximize the variance of the data. PCA reduces dimensionality while preserving data structure and large-scale relative distances. In our case, the resulting axes can be interpreted as linear combinations of LCA components.

On the other hand, the t-Distributed Stochastic Neighbor Embedding (t-SNE) method is nonlinear and probabilistic. This projection method calculates the probability that two points are neighbors in high-dimensional space and attempts to reproduce this probability in 2D/3D. Mathematically, this involves minimizing the Kullback-Leibler divergence. This tool is useful for identifying clusters that are invisible in PCA, even though it distorts global distances.

For studying the global structure of latent space, PCA is preferred. However, for embedding discrete variables, it can be useful to analyze the latent space using the t-SNE projection if clusters are not visible with PCA. In conclusion, a combination of these projection methods can be used for a more in-depth visualization of the latent space.

4.4 Results on 8 Indicators

These 8-dimensional points are projected into spaces of respective dimensions 3 (GWP, LOP, WCP) and 2 (GWP, WCP) to facilitate visualization. It is important to keep in mind that the best-performing solutions according to the LCA methodology are those located close to the origin. The distribution of the generated solutions in 3D space provides a good indication of the model's behavior. The least efficient models, or those whose training shows prior collapse, generate a limited diversity of structures (located at a few points or lines in the LCA value space), often far from the targets.

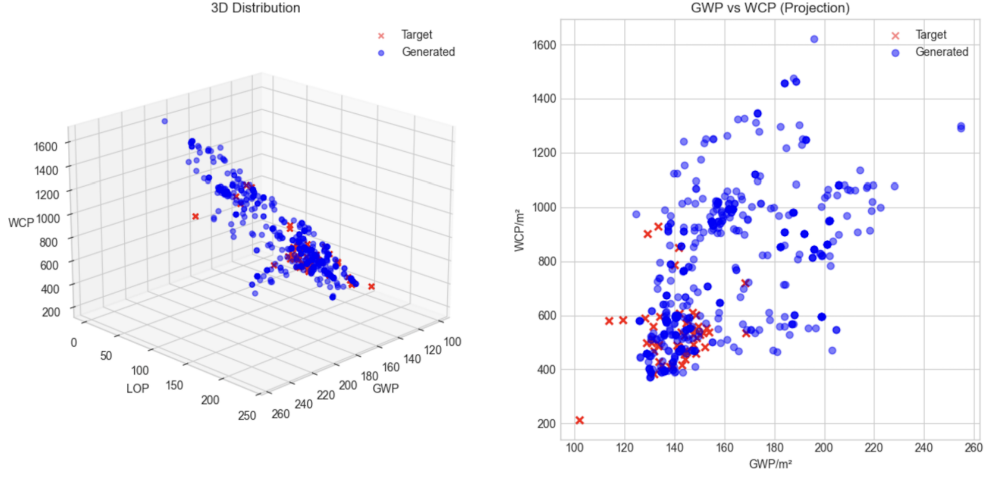


Fig. 5 $n_{targets} \times p$ structures generated by the model $m = 13$ (blue dots). The $n_{targets}$ targets are represented by red crosses.

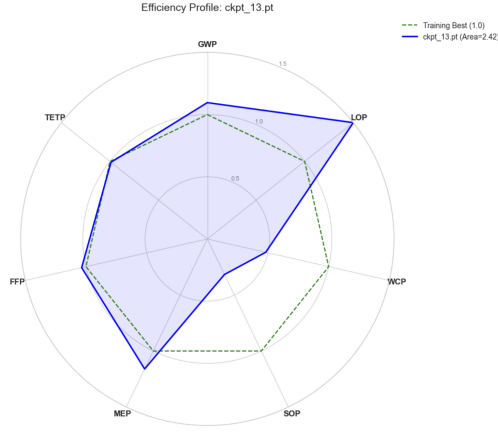


Fig. 6 Kiviati diagram for a model comparing the normalized LCA indicators of a generated solution with the indicators from the training dataset.

Therefore, measuring the hypervolume of generated solutions is a relevant metric for model evaluation. To measure hypervolume, it strictly reflects the dominance order, as defined by Life Cycle Assessment (LCA), of the designs generated by the models.

It is also possible to represent, for each dimension of the LCA vector, the performance of a solution generated by the model against the data in the training dataset. In the Kiviati diagram, the performance of a dominant solution from the model is shown in blue, while the optimum in the training dataset for each component of the LCA/area vector is shown in green. The axes of the diagram are normalized to these values.

4.4.1 Validation of the architectural choice

Hypervolume analysis allows us to draw conclusions from the approach used. The generative models studied are able to reproduce a significant fraction of the surface-normalized Pareto hypervolume of the training dataset, with 79.2% for the best model according to this metric ($m = 84$). This indicator reveals the CVAE’s ability to address the LCA problem by offering a single model that solves various LCA problems by summarizing their characteristics in the latent space. Furthermore, the analysis of Kiviat diagrams shows that while the best models do not manage to dominate the best structures of the training dataset, they are able to surpass performance on certain indicators. This validates the CVAE architecture for providing novel solutions to LCA challenges.

It is important to clarify that this metric does not take into account the average error on the generated surface (geometric condition), nor the visible structuring of the latent space, nor the diversity of the generated structures.

4.4.2 Case study of a latent space for an 8-indicator model

In this section, a case study is conducted on a high-performing model whose latent space exhibits a manifest organization. The objective is to determine whether this latent space compresses the training dataset, which is enriched by grouping according to the various LCA-Structure trade-offs. Following training, the latent space, with a dimension of 16, is projected to a dimension of 2 using principal component analysis to enable visualization. In Figure 7, this same space is then colored according to the gradients of the 8 LCA indicators, and subsequently, in Figure 7, colored according to the parameters of the associated structures.

In the Figure 8 representing the input space, a structure of distinct, thread-like clusters is observed. The encoder has clearly differentiated the data according to the materials of the columns and beams, with gradients within each cluster based on the floor material. Then, Nb floors, Building width, and Building length create the vertical dispersion within each cluster. The orientation of the thread-like clusters is dictated by the stair distance, with low values serving as points of contact for the branches. Gradients of building length and building width are also observed within each cluster. The strong clustering of the latent space according to the building characteristics was expected, given the nature of the training dataset, which contains a majority of discrete variables.

We observe that the majority group of positively correlated indicators (GWP, WCP, MEP, FFP, TETP) exhibits gradients oriented in the same direction within the clusters. Conversely, pairs of uncorrelated indicators (GWP - LOP, OPD - WCP) show gradients in opposite directions. Visual analysis confirms that environmental performance is dominated by two forces: geometric compactness and material choice.

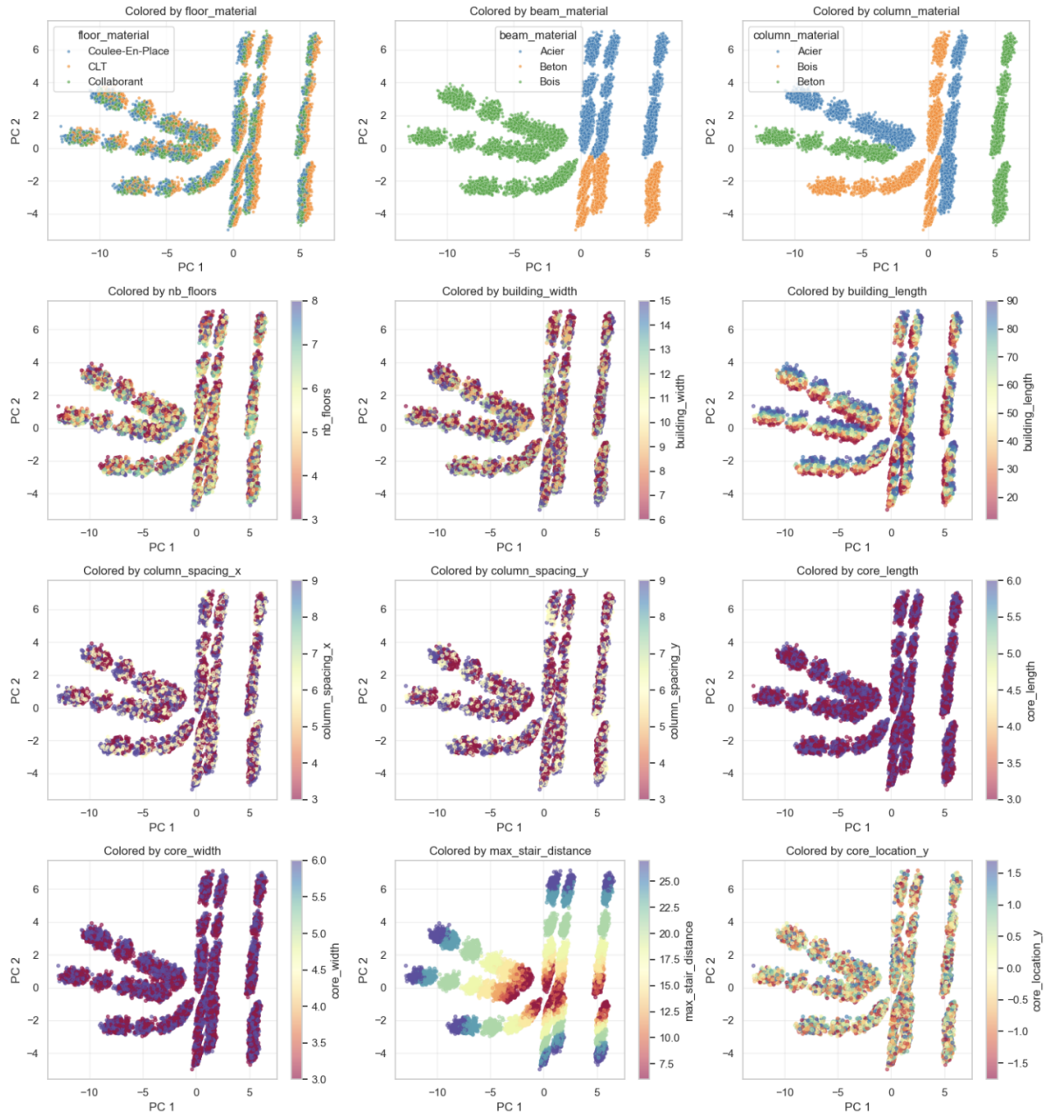


Fig. 7 Visualization of the latent space of the model $m = 13$, colored according to the values of the LCA indicators.

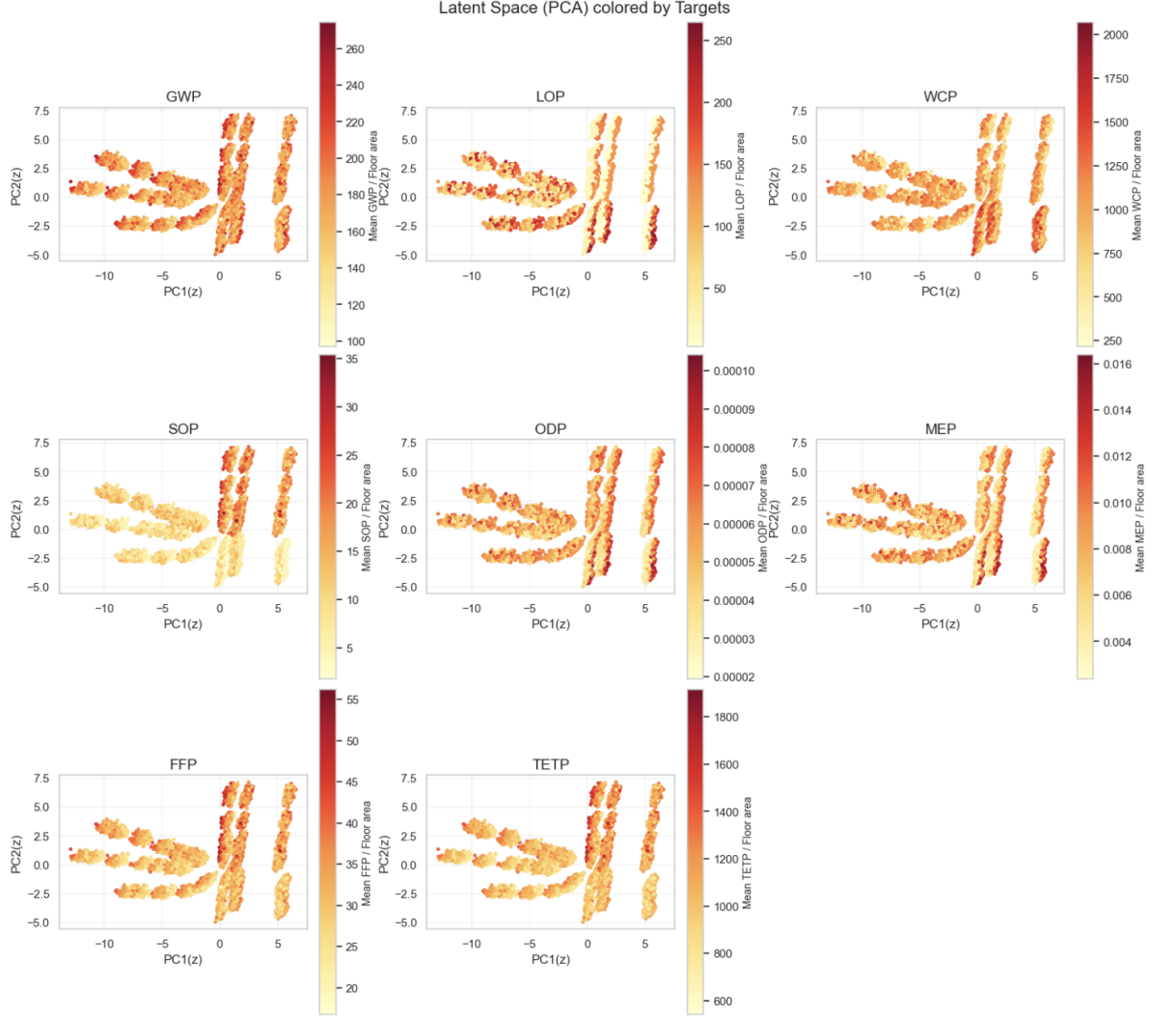


Fig. 8 Visualization of the latent space of the model $m = 13$, colored according to the values of the building features.

5 Advanced Modeling: Handling Correlations

This section proposes methods for improving model architecture, as well as ways to better account for the specific characteristics of the LCA methodology.

As established in the preceding sections of this report, the study of structure generation under LCA constraints requires consideration of the specific characteristics of the LCA methodology. These characteristics also played a role in training statistical learning models. LCA indicators have *a priori* very different orders of magnitude, as

well as for the evaluation and visualization of results, since midpoint LCA indicators are not directly comparable to each other.

To further explore the representation of the characteristics of LCA indicators in a CVAE-type machine learning model, it is necessary to consider how the correlations of these indicators are learned by the model. As such, as has been established, the latent space of a VAE or CVAE allows for the representation of LCA-Structure trade-offs while respecting the correlations between indicators. However, it may be possible to further improve model performance by carefully selecting model characteristics. This part of this work seeks to answer this question.

5.1 Structuring the latent space with Mahalanobis distance

The following equation defines the conditional prior as a Gaussian distribution with diagonal covariance:

$$p_\psi(z|y) = \mathcal{N}(\mu_p(y), \text{diag}(\sigma_p^2(y)))$$

This formulation forces the latent dimensions to be independent of each other, preventing the direct modeling of correlations between multiple LCA indicators. To capture the correlations, we replace the diagonal variance with a Full Covariance Matrix. The network does not directly predict (to ensure it is positive definite), but its Cholesky factor does. $\Sigma_p(y) = L_p(y)L_p(y)^T$

The prior now becomes : $p_\psi(z|y) = \mathcal{N}(\mu_p(y), \Sigma_p(y))$. And the likelihood that the model tries to maximize is:

$$p_\psi(z|y) = \frac{1}{(2\pi)^{k/2} |\Sigma_p(y)|^{1/2}} \exp \left(-\frac{1}{2} (z - \mu_p(y))^T \Sigma_p(y)^{-1} (z - \mu_p(y)) \right)$$

By taking the logarithm to obtain the log-likelihood (maximized term in the ELBO), the Mahalanobis distance D_M appears explicitly:

$$\ln p_\psi(z|y) = -\frac{1}{2} \underbrace{(z - \mu_p(y))^T \Sigma_p(y)^{-1} (z - \mu_p(y))}_{D_M^2(z, \mu_p)} - \frac{1}{2} \ln |\Sigma_p(y)| + C$$

The integration of this architectural modification was proposed and tested on a limited number of models. Performance appears to be comparable in diversity and effectiveness to standard models. The results of this study are not analyzed in this work.

5.2 Dimension of the latent space

By allowing the estimation of the dimensionality of the training dataset, the Levina-Bickel estimator is a topological indicator that enables us to determine the minimum number of dimensions needed in the latent space to represent the diversity of the data. The density of nearest neighbors is analyzed around each point. If the volume

increases slowly with distance, the intrinsic dimension is small, and *vice versa*. The value of the Levina-Bickel estimator [4] can help define the dimensionality of the latent space, providing insight into the trade-off between the depth of the latent space (which improves generator performance) and its shallowness (which increases the chances of successful visualization). The dimension of the embedding is essential: if the dimension is too small, important data features are “collapsed” onto the same dimension, and if the dimension is too large, the projections become noisy and, in some cases, unstable.

In the training dataset, the intrinsic dimensionality of the data is estimated to be 16 dimensions. Therefore, a minimum of 16 dimensions in the latent space is required to represent the diversity of the data. Furthermore, another approach involves not only studying the dimensionality of the latent space but also forcing certain chosen parameters to be represented “by default” in specific principal components of the latent space.

6 Discussion

In conclusion, we can say that the objectives set for this work have been met. The proposed architectures successfully address both the forward and inverse problems using a database of structures and even generate structures with better environmental performance than those in the training database, based on a limited number of indicators. Beyond the performance of the generated structures, this work has also led to reflection on how to integrate LCA into a deep learning methodology in terms of model evaluation, parameter and architecture selection, and results visualization. In particular, the latent space can be used as a representation of LCA-Structure trade-offs, potentially leading to an interactive tool for navigating the latent space while generating structures in real time. However, optimizing the latent space must not compromise the decoder’s performance.

Another approach to evaluating structures generated by a deep learning model involves adapting sampling methods to threshold-based standards. Most environmental regulations set thresholds for environmental indicator values. Thus, the validity of a generated structure is based on this threshold. A multi-criteria threshold can be represented by a hyperplane in the LCA objective space. It would be beneficial to propose sampling methods based on these thresholds rather than fixed values.

Ultimately, the value of these models will be fully realized with a larger and more diverse database. It is also essential to consider the energy performance of buildings, particularly a thermal study. To achieve this, it is necessary to successfully involve stakeholders in the construction industry in a process that could—in the long run—make architecture more sustainable.

References

- [1] Yiyang Huang et al. “Performance-Driven Generative Design in Buildings: A Systematic Review”. In: *Buildings* 15.24 (2025). DOI: [10.3390/buildings15244556](https://doi.org/10.3390/buildings15244556).
- [2] Olivier Jolliet et al. *Analyse du cycle de vie: Comprendre et réaliser un écobilan*. 3rd ed. Lausanne, Suisse: Presses polytechniques et universitaires romandes (PPUR), 2017. ISBN: 978-2-88915-200-2.

- [3] Diederik P Kingma and Max Welling. “Auto-Encoding Variational Bayes”. In: *International Conference on Learning Representations (ICLR)*. 2014. URL: <https://arxiv.org/abs/1312.6114>.
- [4] E. Levina and P. J. Bickel. “Maximum likelihood estimation of intrinsic dimension”. In: *Advances in Neural Information Processing Systems* 17 (2004).
- [5] P. Navaro-Auburtin. *Life Cycle Assessment for the design of building structures : from materials to structural*. Lieu de publication: Nom de l’éditeur ou de l’université, 2025.
- [6] Ali Nouri and Ming Hu. “Democratizing environmental impact assessment: An AI-augmented framework for sustainable building design”. In: *iScience* 29.2 (Dec. 2025). PMID: PMC12856274, p. 114589. DOI: [10.1016/j.isci.2025.114589](https://doi.org/10.1016/j.isci.2025.114589).
- [7] Maxime Pollet, Olivier Baverel, and Adélaïde Feraille. “Life Cycle Analysis-Informed Conceptual Design of Building Structures Using Deep Generative Models”. In: *Proceedings of ARTISTE 2025*. Laboratoire Navier, ENPC, Université Gustave Eiffel, CNRS. France, 2025.
- [8] Kihyuk Sohn, Honglak Lee, and Xinchen Yan. “Learning Structured Output Representation using Deep Conditional Generative Models”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 28. 2015, pp. 3483–3491. URL: <https://papers.nips.cc/paper/5775-learning-structured-output-representation-using-deep-conditional-generative-models>.